

WASKESIU LAKE, SK, Canada

WMO: 714540

Lat: 53.920N

Lon: 106.070W

Elev: 1868

StdP: 13.73

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-22.7	-34.2	0.8	-27.1	-29.3	1.1	-22.6	15.4	14.9	13.7	13.8	3.4	310	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	81.5	63.7	78.5	62.2	75.6	60.5	66.4	77.3	64.4	74.6	62.7	71.9	6.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.6	91.1	69.7	60.8	85.2	68.3	59.0	79.9	66.7	32.1	77.7	30.6	74.7	29.3	72.0	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.7	12.0	10.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-32.6	87.9	4.6	3.1	-35.9	90.2	-38.5	92.0	-41.1	93.7	-44.4	95.9
			-32.8	70.1	4.6	1.8	-36.0	71.4	-38.7	72.5	-41.3	73.5	-44.6	74.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	3.5	7.9	20.1	35.7	48.7	58.3	63.4	61.2	51.7	37.8	20.6	7.4	(6)
(7)		DBStd	23.89	15.75	13.97	14.01	10.61	8.85	6.53	5.31	6.13	7.94	8.54	11.80	14.52	(7)
(8)		HDD50	6798	1441	1178	928	440	133	11	0	2	73	389	883	1320	(8)
(9)		HDD65	11103	1906	1598	1393	879	507	214	93	149	402	844	1333	1785	(9)
(10)		CDD50	1265	0	0	0	10	93	261	417	350	124	10	0	0	(10)
(11)		CDD65	95	0	0	0	0	2	14	45	31	3	0	0	0	(11)
(12)		CDH74	962	0	0	0	2	54	161	397	294	53	1	0	0	(12)
(13)		CDH80	160	0	0	0	0	10	27	65	52	6	0	0	0	(13)
(14)	Wind	WSAvg	5.0	4.6	4.7	5.3	5.7	5.4	4.8	4.6	4.6	4.9	5.6	5.3	4.7	(14)
(15)	Precipitation	PrecAvg	18.8	0.7	0.6	0.7	1.3	1.7	3.5	3.1	2.3	1.7	1.5	0.9	0.8	(15)
(16)		PrecMax	24.5	1.2	0.9	1.5	3.1	5.0	9.4	5.9	4.6	6.5	4.6	2.2	3.3	(16)
(17)		PrecMin	10.9	0.1	0.1	0.0	0.2	0.1	1.6	1.7	0.6	0.0	0.2	0.0	0.1	(17)
(18)		PrecStd	3.7	0.2	0.3	0.4	0.9	1.2	1.6	1.2	1.1	1.4	1.1	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	43.7	53.7	69.8	80.2	83.4	85.8	85.5	80.0	68.5	49.5	40.4	(19)
MCWB			35.8	35.7	41.1	49.7	57.9	62.8	66.2	65.6	61.2	53.7	41.0	35.0	(20)	
2%		DB	35.0	37.0	47.4	63.0	73.8	78.4	81.4	80.8	74.2	59.8	43.3	34.9	(21)	
		MCWB	31.8	31.9	37.7	47.1	54.0	60.6	64.9	63.8	58.4	48.6	37.4	31.3	(22)	
5%		DB	30.5	32.2	42.9	57.9	69.5	74.9	78.4	77.2	69.5	54.8	38.4	30.8	(23)	
		MCWB	28.6	29.3	35.0	44.4	51.5	58.6	63.6	62.1	56.3	45.8	34.4	28.6	(24)	
10%		DB	26.1	28.0	38.1	52.6	65.5	71.5	75.4	73.7	64.9	50.3	34.9	26.8	(25)	
		MCWB	24.6	26.0	32.9	41.4	50.0	56.9	61.8	60.8	54.3	43.5	32.2	25.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	36.6	42.0	51.8	59.2	66.7	70.1	68.3	62.9	54.3	41.9	35.7	(27)
MCDWB			41.1	42.4	52.7	65.5	72.5	78.9	80.5	79.7	75.0	66.8	48.5	39.6	(28)	
2%		WB	32.1	32.6	38.3	48.0	56.2	63.3	67.1	65.7	59.9	50.1	37.9	31.8	(29)	
		MCDWB	34.5	36.5	46.0	61.4	69.5	72.6	77.6	77.1	70.8	57.4	42.1	34.5	(30)	
5%		WB	28.5	29.3	35.7	44.8	53.7	61.1	65.1	63.8	57.5	46.7	34.9	28.6	(31)	
		MCDWB	30.4	32.1	41.5	56.3	66.0	70.2	75.0	74.3	66.9	53.5	37.9	30.7	(32)	
10%		WB	24.7	26.1	33.1	42.1	51.3	59.1	63.4	62.0	55.2	44.1	32.3	25.3	(33)	
		MCDWB	26.2	28.1	37.7	51.9	63.2	68.3	72.5	71.3	63.8	49.7	34.8	26.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.9	18.9	19.9	19.8	21.5	18.9	18.5	17.6	14.1	11.9	13.6	(35)	
MCDBR			18.7	21.9	23.0	26.8	27.0	24.7	23.4	23.8	25.6	22.7	16.2	17.1	(36)	
MCWBR			15.6	17.0	14.3	14.2	12.0	10.9	10.3	10.6	12.6	12.9	11.2	14.3	(37)	
5% WB		MCDBR	18.0	21.0	21.4	25.0	24.5	20.1	20.7	21.3	23.0	21.5	15.0	16.4	(38)	
		MCWBR	15.3	16.5	13.9	14.1	11.8	10.4	10.5	10.7	12.2	12.8	11.0	14.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.230	0.236	0.277	0.317	0.339	0.361	0.364	0.356	0.307	0.288	0.248	0.219	(40)	
(41)		taud	2.345	2.378	2.357	2.439	2.437	2.407	2.395	2.430	2.579	2.579	2.471	2.356	(41)	
(42)		Ebn at Noon	246	284	290	291	288	282	278	274	279	261	239	229	(42)	
(43)		Edh at Noon	17	23	30	31	33	35	35	32	24	19	15	13	(43)	
(44)	All-Sky Solar Radiation	RadAvg	289	622	1112	1523	1790	1792	1825	1521	1053	599	268	198	(44)	
(45)		RadStd	29	36	79	128	159	132	127	129	119	96	42	23	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	-0.61	-0.83	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page